

EXAM 7 – SPRING 2014

18. (4 points)

a. (0.5 point)

Define market risk and credit risk.

b. (1.5 points)

Identify and briefly describe three types of market risk that an insurer may face.

c. (1 point)

Briefly describe how market risk and credit risk each affect the value of both invested assets and insurance liabilities.

d. (1 point)

Given the following amounts (in \$000,000) for each risk factor:

Market risk of assets:	100
Credit risk of assets:	40
Market risk of liabilities:	40
Credit risk of liabilities:	10

Assuming non-negative correlations, calculate a range for total risk from these factors.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 18	
TOTAL POINT VALUE: 4	LEARNING OBJECTIVE: C1: Demonstrate how insurance and financial risk can be analyzed quantitatively. C4: Demonstrate the properties of various risk measures and their limitations.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answer 1</i> Market Risk is the risk of loss due to changing asset and liability values as a result of changing market prices. Credit risk is the risk of loss due to a change in credit quality (downgrade or default) of a counterparty.	
<i>Sample Answer 2</i> Market risk – the risk that changes in financial variables such as interest rate will change the value of assets and liabilities. Credit risk – the risk that a party will be unable to fulfill its financial obligations.	
Part b: 1.5 points	
<i>Sample Answer 1</i> Concentration Risk – lots of assets w/same issuers (think 100% Disney bonds) or same quality (all BBB); Reinvestment Risk – risk that when principal is paid, will not earn target return on new money (i.e. bonds with 6% yield mature and now rate for bonds of same quality is 2%; Equity/Property Risk – Risk of change or fluctuations in value of stocks & property (stock market crash '08 and housing crisis)	
<i>Sample Answer 2</i> Interest Rate Risk – the risk due to fluctuation of interest rates; Basis risk – the risk due to yields on different debt instruments not moving together; Assets/Liabilities Mismatch Risk – the risk due to the mismatch of the timing of the assets cash flow and the liabilities cash flow.	
Part c: 1 point	
<i>Sample Answer 1</i> Market/Asset – market risk due to reduction in value of financial instruments (e.g. equity); Market/Liability – market risk if insurance liability is discounted (e.g. using risk-free rate), then movement in risk-free rate would impact liabilities; Credit/Asset – credit risk could affect value of corporate/government bonds if the issuer defaults or gets downgraded; Credit/Liability –reinsurance recoverables may not be recoverable if reinsurer defaults, thus increasing net liability.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

Market risk affects stock prices and bond prices, which are assets held, but also affects liabilities through inflation. Credit risk affects bonds held (will company forego coupon payment or default on all obligations) but also affects liabilities in that insureds may not be able to pay the portion ceded to them (primary insurer is then liable).

Part d: 1 point

Sample Answer 1

Assuming complete independence (0% correlation): $\sigma^2_{\text{total}} = \sum \sigma^2_{\text{factor}}$ so total risk = $(100^2 + 40^2 + 40^2 + 10^2)^{0.5} = 115.326$. Assuming 100% correlation: $\sigma^2_{\text{total}} = (\sum \sigma_{\text{factor}})^2$ so total risk = $[(100 + 40 + 40 + 10)^2]^{0.5} = 190$. Total range = [115.326, 190]

Sample Answer 2

The range for total risk would be from zero correlation to perfect correlation. Min risk = $(100^2 + 40^2 + 40^2 + 10^2)^{0.5} = 115.326$. Max risk = $100 + 40 + 40 + 10 = 190$. Range is (115.3, 190)

Sample Answer 3

Total independent = $(100^2 + 40^2 + 40^2 + 10^2)^{0.5} = \115.33M . Total dependence = $100 + 40 + 40 + 10 = \$190\text{M}$. Between (\$115.33M, \$190M)

EXAMINER'S REPORT

Part a

The candidate was expected to be able to describe types of insurer risks. Most candidates received full credit and those that didn't usually stated that there was a risk that the value of assets and liabilities would change but failed to provide a reason for the change (e.g., financial variables).

Part b

The candidate was expected to be able to describe types of market risk. The majority of candidates received full credit. Those that lost partial credit did so because they used examples of another insurer risk (e.g., credit risk, underwriting risk), forgot to identify the market risk, or duplicated the description of Equity/Property risk.

Part c

This question required the candidate to take a learning objective and be able to apply it with a real world example. The candidate was expected to provide four specific items: an example of both an asset and a liability, and a description of the direct impact from both a market risk and a credit risk. Although many candidates were able to use their answer from part b. to help identify a real world example, other candidates simply stated that market risk would change the value of assets and liabilities but failed to provide an actual example or reason why they would change. Another common error was a candidate citing asset/liability mismatch risk but this was not an acceptable answer as it does not have a direct impact to the value of liabilities.

Part d

The candidate was expected to recognize that the maximum risk assumes full

SAMPLE ANSWERS AND EXAMINER'S REPORT

correlation/dependency between all of the risks and that the minimum risk would occur when the risks have no correlation or are independent of each other. Most candidates had some difficulty with this question. One common error was a candidate using the more conservative equation IAA provided in Appendix E, section 8.18, where IAA specifically assumed the best diversification can be approximated by a correlation coefficient of 0.5. Another common error was a candidate offsetting the risks which was not a valid approach since the question clearly stated "assuming non-negative correlations".